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MCP (Model Context Protocol) Quiz

Total Questions: 20

Type: MCQ + Scenario-Based + Short Answer

Difficulty: Medium to Advanced



Section A: Multiple Choice Questions (MCQs)

Q1. What is the primary purpose of the Model Context Protocol (MCP)?

- A. Data storage
 - B. Standardized agent-tool communication
 - C. Model training
 - D. UI rendering
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Q2. Which MCP integration pattern focuses on delegating tasks across multiple agents?

- A. Single-agent pattern
 - B. Pipeline pattern
 - C. Multi-agent orchestration
 - D. Batch processing
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Q3. In MCP, tool invocation typically follows which structure?

- A. Direct function call
 - B. API request with structured schema
 - C. Manual input
 - D. File upload
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Q4. Which component ensures that agents can safely share data?

- A. Tokenization
- B. Secure context sharing

- C. Logging
 - D. Serialization
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Q5. What is a key advantage of MCP in enterprise environments?

- A. Faster UI rendering
 - B. Vendor lock-in
 - C. Interoperability across tools
 - D. Reduced storage
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Q6. Which integration pattern is best suited for sequential task execution?

- A. Event-driven pattern
 - B. Pipeline pattern
 - C. Mesh pattern
 - D. Parallel pattern
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Q7. What does “context” refer to in MCP?

- A. UI design
 - B. Model weights
 - C. Data passed between agents/tools
 - D. Network latency
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Q8. Which of the following improves security in MCP?

- A. Hardcoding credentials
 - B. Context encryption
 - C. Open access APIs
 - D. Public tokens
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Section B: Scenario-Based Questions

Q9. A financial company uses multiple AI agents for fraud detection, reporting, and alerting. Which MCP pattern is most suitable?

- A. Single-agent
- B. Multi-agent orchestration

- C. Static pipeline
 - D. Batch processing
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Q10. An AI system fetches stock data, analyzes it, and generates a report in sequence.

Which pattern is being used?

- A. Parallel processing
 - B. Pipeline pattern**
 - C. Mesh network
 - D. Event loop
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Q11. An agent calls an external weather API using MCP.

What is this process called?

- A. Context sharing
 - B. Tool invocation**
 - C. Model tuning
 - D. Serialization
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Q12. A healthcare system ensures patient data is encrypted before sharing between agents.

Which concept is applied?

- A. Data batching
 - B. Secure context sharing**
 - C. Logging
 - D. Caching
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Q13. A chatbot uses MCP to connect with CRM, database, and analytics tools.

What is this an example of?

- A. Standalone AI
 - B. Enterprise agent integration**
 - C. Local processing
 - D. Static system
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Section C: Short Answer Questions

Q14. Explain the difference between *pipeline pattern* and *multi-agent orchestration* in MCP.

In Pipeline the tasks done in sequence (step-by-step).
But Multi-agent orchestration the multiple agents collaborate dynamically.

Q15. Describe how tool invocation works in MCP architecture.

Agent sends structured request → MCP routes it → tool/API executes → response returned.

Q16. What are the risks of insecure context sharing? Mention at least two.

- ☐ Data leakage
 - ☐ Unauthorized access
 - ☐ Privacy violations
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Q17. How does MCP improve scalability in enterprise AI systems?

MCP enables modular agents + tools → easy scaling by adding more agents/services.

Section D: Case Study / Analytical

Q18.

A retail company wants to build an AI system that:

- Retrieves customer data
- Recommends products
- Sends personalized emails

Question:

Design an MCP-based integration approach and justify your chosen pattern.

Approach: Use **Pipeline Pattern with MCP**

Flow:

1. Agent 1 → retrieves customer data (CRM/database)
2. Agent 2 → generates product recommendations (ML model)
3. Agent 3 → sends personalized emails (email service)

We use Pipeline because:

- Tasks are **sequential and dependent**
 - Output of one step → input to next
 - Easy to manage and scale
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Q19.

An organization is concerned about data privacy when multiple agents interact.

Question:

Suggest strategies using MCP to ensure secure context sharing.

Strategies using MCP:

Encryption → protect data during transfer

Role-Based Access Control (RBAC) → limit who can access what

Authentication & Authorization → verify agents/tools

Audit Logs → track data usage

Data minimization → share only required data

Result: Prevents leaks, ensures privacy, and maintains compliance

Q20.

Compare MCP-based architecture with traditional API-based integration in enterprise systems.

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Result: Prevents leaks, ensures privacy, and maintains compliance
